MIL-P-60523A (AR)
AMENDMENT 2
12 April 1981
SUPERSEDING
AMENDMENT 1
28 October 1980

MILITARY SPECIFICATION

PROJECTILE, 165MM, HEP
M123A1, METAL PARTS

This amendment forms a part of Military Specification MIL-P-60523A, dated 18 May 1979.

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Paragraph 3.5.1, delete and substitute the following: "3.5.1 Raw material. The billets, bars, tubes or plate material from which projectile components are formed shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, folds, bursts, seams, laminations and other metal defects."

ADD:

Paragraph 3.1.1 - Rotating Band Application - The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawing and specifications. Gas metal arc welding process in this case means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained oscillating the welding head and the auxiliary wire feed system. The process for applying the rotating band, once established must be maintained, controlled and monitored by the contractor. Procedures for mantaining controlling, and monitoring the process must be submitted to the technical agency for approval prior to start of production. Documented records of the process control must be made available to the Government upon request.

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Paragraph 3.1.2 - Foreign Matter on Band Seat - Prior to banding the band seat should be free of any contaminants that would be detrimental to the adhesion of the band to the band seat.

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Paragraph 3.6.b, delete "band (s)" and substitute "band"

Paragraph 3.6.c, delete "band (s)" and substitute "band"

Paragraph 3.8.2, add to end: "see (6.8)."

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ADD:

To TABLE I:

EXAMINATION OR TEST COLUMN - BODY ASSEMBLY PRIOR TO PAINTING (DWG 9327406)"

"MAGNETIC PARTICLE INSPECTION"

No. of sample units "5"
Requirement Paragraph - "3.1.1"
Paragraph Reference/Inspection Method - "4.5.6.4.1"

"Band Chemical Analysis"
No. of Sample Units "1"
Requirement Paragraph - "3.1.1
Paragraph Reference/Inspection Method - "4.5.6.4.2"

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Paragraph 4.4.2.1.3, delete and substitute the following: "4.4.2.1.3 Inspections of raw material prior to forming.- Each slug, tube or disc obtained from bar, tube or plate stock shall be visually examined for surface indications of the types of metal defects described in 3.5.1 which are visible to the unaided eye on all exterior surfaces. Visual standards may be selected to aid in the determination of a slug's, tube's or disc's acceptability."

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Paragraph 4.4.2.2, delete "Critical Defect number 2" it its entirety.

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Paragraph 4.4.2.3, delete "Critical Defect number 1" in its entirety and substitute: "None defined"

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Paragraph 4.4.2.4, add the following Critical Defects and Note:

"4.' Magnetic particle inspection (Note 1), Requirement paragraph, '3.1', Paragraph Reference/Inspection Method, 'Test/4.5.6.'"

"5.' Magnetic particle inspection acceptance stamp missing,' Requirement paragraph, '3.1,' Paragraph Reference/Inspection Method,' Visual.'"

"Note 1: Magnetic particle inspection shall be performed on projectile bodies which have previously been subjected to the hydrostatic and air pressure tests."

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ADD:

TO PARAGRAPH 4.4.2.4

MAJOR: "127 Magnetic particle inspection of bandseat"
No. of sample units - "4.4.3.4.1"
Requirement paragraph - "3.1.1"
Paragraph Reference/Inspection Method - "4.5.6.4.2"

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Paragraph 4.4.2.7, Title, delete "Carton, Packing, Prior to Sealing" and substitute "Carton, Packed, Prior to Sealing (Base Plug)"

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Paragraph 4.4.2.8, Title, delete "Cartons, Packed, After Sealing" and substitute "Cartons, Packed, After Sealing (Base Plug)"

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Paragraph 4.4.2.9, Title, delete "Pallet" and substitute "Pallet (Projectile Body)"

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Paragraph 4.4.3.1.1, Delete third sentence and substitute the following:

"If two or more projectile bodies of a sample fail to meet the hardness requirement specified on dwg. 9327406, the batch represented by that sample shall be rejected and a new stress-relieved lot established as prescribed herein. The first batch of the new stress-relieved lot may be a failing batch which, after re-processing and subsequent testing, meets the applicable requirements."

Paragraph 4.4.3.1.2, Delete and substitute the following:

"4.4.3.1.2 Tensile strength and elongation of bourrelet.- Two projectile bodies of the twenty-five bodies selected from the first batch of a stress-relieved lot for the hardness test of 4.4.3.1.1 shall be subjected to this test. Following the satisfactory compliance of this initial sample with the requirements of dwg. 8845046, a sample of two projectile bodies from every five batches, or fraction thereof, of a stress-relieved lot shall be selected for this test. Failure of any projectile body tensile specimen to meet the applicable requirements shall reject the lot of stress-relieved bodies which it represents and shall require the establishment of a new stress-relieved lot as prescribed in 4.4.3.1.1."

Paragraph 4.4.3.2, delete the second sentence and substitute the following:

"Failure to comply with the applicable specification and drawing requirements shall be cause for the rejection of the meterial represented by the failing samples."

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Paragraph 4.5.1.1, delete and substitute the following:

"4.5.1.1 Hardness of the nose/ogive.- After final stress relief, but prior to final machining, a Rockwell Superficial Hardness Test shall be performed on the samples provided by 4.4.3.1.1. The requirement and procedures for performing this test are prescribed by ASTM E18. Four equally spaced readings shall be taken at a point which is four inches from the tip of the projectile body nose and averaged in order to determine the acceptability of each sample.

One reading shall be taken on the inside surface of the projectile body tensile samples, obtained as prescribed in 4.5.1.2, at locations directly opposite points that are one, four and seven inches from the projectile body nose, for information only. Two equally spaced and averaged readings on each sample shall be taken at the tip of the projectile body nose when the nose is welded or brazed as allowed by dwg. 8845046."

ADD:"

Paragraph 4.4.3.4 - Rotating Band

Paragraph 4.4.3.4.1 - Magnetic Particle Inspection of Band Seat - The first five banded bodies after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test to determine compliance with the visual standards set by the responsible technical engineering agency for that area of the body defined on dwgs 9327406 and 8845046 (defined as band seat area). Acceptance criteria specified herein applies separately for each welding machine in operation. Reduced sampling of one in twenty band bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been visually examined and found acceptable. Reduced sampling of one in each 8 hours production per welding machine operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been visually examined and found acceptable. (See 6.10)

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Paragraph 4.4.3.4.2 - Band Chemical Analysis - One body after band from each 4 hours production or fraction thereof for each welding machine in operation shall be subjected to this test. Failure to comply with requirements of the drawing shall be cause for rejection of the production represented by the sample.

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Paragraph 4.5.1.4, Add the following title: "Reporting and format of data."

Paragraph 4.5.2, Add the following sentence:

"The contractor shall prepare a written standard operation procedure, to include the inspection of projectile bodies and calibration procedures, which shall be submitted for the review and approval of the Contracting Officer prior to the submission of the first article sample (see 6.2)"

Paragraph 4.5.2.1, Delete the last sentence in its entirety.

Paragraph 4.5.3 - delete the third sentence and rewrite as follows, "The body shall be mounted in an approved fixture in a manner such that atleast fifty percent of the body from the nose end shall be submerged in water."

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Paragraph 4.5.3, Add the following sentence:

"The contractor shall prepare a written standard operation procedure, to include the inspection of projectile bodies and calibration procedures, which shall be submitted for the review and approval of the Contracting Officer prior to the submission of the first article sample (see 6.2)."

Paragraph 4.5.3.1, Delete this paragraph in its entirety.

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Paragraph 4.5.5.4, Add the following sentence:

"The proving ground shall prepare a written assembly procedure for the assembly of projectile metal parts ballistic samples and submit the procedure for review to the Contracting Officer prior to the assembly of the first article ballistic test samples (see 6.4)."

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ADD:

Paragraph 4.5.6.4,- Rotating Band

Paragraph 4.5.6.4.1 - Magnetic Particle Inspection of Band Seat - samples provided in accordance with 4.4.3.4.1 after heat treatment shall have the band removed by machining the band and chemically etching without damaging the base metal (steel) of the band seat (see 6.9). The procedure for performing this test shall be specified in MIL-I-6868E for wet fluorescent method, inducing a circular magnetic field, followed by longitudinal magnetic field.

Paragraph 4.5.6.4.2 - Band Chemical Analysis - From the bodies selected in 4.4.5.4.2 sample chips shall be obtained from each band from the last .030 inches removed by machining prior to finish diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet.

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Paragraph 5.2.2.2, Delete and substitute the following:

"5.2.2.2 Size.-Dimensions of the fiberboard boxes shall be 5 1/2" W X 5 1/2" L X 12 3/16" D. The dimension of the discs shall be 5 inches in diameter."

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Add the following paragraphs:

"6.8 Burr.-A burr that is loose, injurious to personnel, interferes with assembly of other components or interferes with inspection gaging is not permissible."

Paragraph 6.9 - Suggested Etching Procedures for Removing Copper Overlay -

A. Recommended: Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay later 0.005 to 0.010 inch thick will dissolve in 8 to 12 hours. Renew solution as necessary.

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B. Solution: 500 grams chromic oxide CR_2O_3 (commercial flakes)

50 ML Suplphuric Acid H_2SO_4 (technical grade)

950 ML Water H_2O

This solution may be used by passing current through the solution (50 amperes DC) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.005 to 0.010 inch thick will require approximately two to three hours to dissolve in fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

Paragraph 6.10 - Alternate Band Seat Samples -The contractor may in lieu of full bodies use rings dimensionally representative of the rear of the M123 body forging at the welding operation and constructed of material indentical to that of the body to fulfill the requirements of Paragraph 4.4.3.4.1. Rings made from scrap shell are acceptable, however, the ring must not have experienced a previous overlay or heat treatment. If a contractor elects to use rings, the procedure for using them must be demonstrated to the technical agency and approval must be obtained. The demonstration must be based on a comparison of the quality and appearance of the overlayed rings versus complete shell.

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